EXPERIMENT-10

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Create a data visualization(e.g.,piecharts,bar graphs) for an Student Course Enrollment System using javascript

Aim:

The aim is to create data visualizations, such as pie charts and bar graphs, for an student course enrollment system using JavaScript.

Procedure:

1. Create an HTML file (index.html) to design the webpage.
2. Add two canvas elements to display bar chart and pie chart.
3. Include the Chart.js library in the HTML file.
4. Create a JavaScript file (app.js) to store data.
5. Define data such as course names and number of students.
6. Write code to generate bar chart and pie chart.

7. Link the JavaScript file with the HTML file.
8. Run the HTML file in a web browser.

Program:

1. index.html:

```html
<!doctype html>
<html>
  <head>
    <title>Course Enrollment Overview</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <link rel="stylesheet" href="style.css" />
  </head>
  <body>
    <div class="container">
      <h1>Course Enrollment Overview</h1>

      <div class="charts">
        <canvas id="barChart"></canvas>
        <canvas id="pieChart"></canvas>
      </div>
    </div>

    <script src="script.js"></script>
  </body>
</html>
```

2.app.js

style.css > canvas

C:\Users\Asus\OneDrive\pie chart\style.css

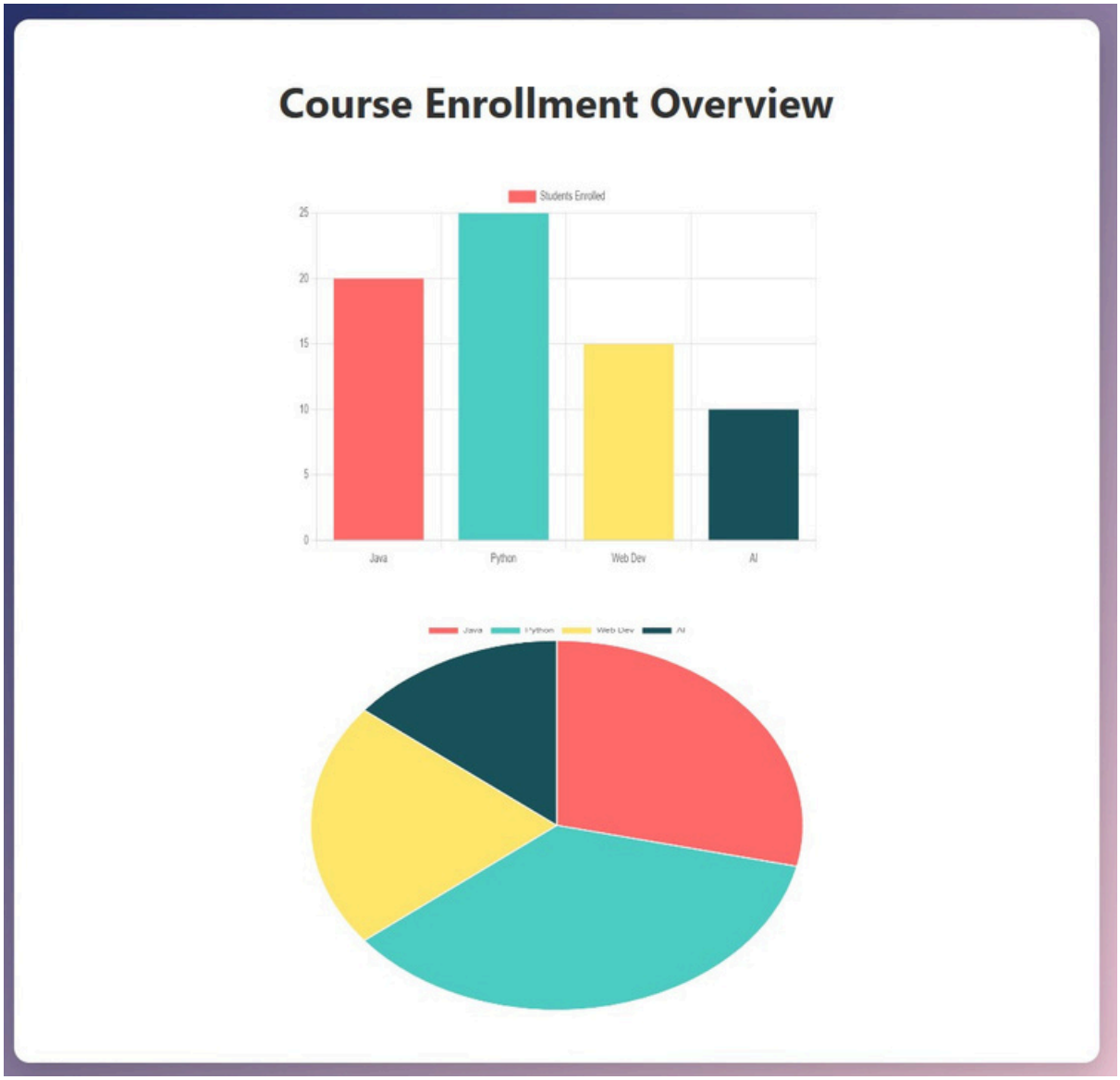
```
  font-family: "Segoe UI", sans-serif;
  background: linear-gradient(135deg, #1d2b64, #f8cdda);
  margin: 0;
  padding: 20px;
  text-align: center;
}

.container {
  background: white;
  padding: 20px;
  border-radius: 12px;
  width: 80%;
  margin: auto;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.2);
}

h1 {
  margin-bottom: 20px;
  color: #333;
}

.charts {
  display: flex;
  justify-content: space-around;
  flex-wrap: wrap;
}

canvas {
  width: 400px !important;
  height: 300px !important;
  margin: 20px;
}
```

3.Script.js

```
JS script.js > data > datasets
C:\Users\Asus\OneDrive\pie chart\script.js
1  n", "Web Dev", "AI"];
2  const dataValues = [20, 25, 15, 10];
3
4  // 🎨 New color theme
5  const colors = [
6    "#ff6b6b", // red
7    "#4ecdc4", // teal
8    "#ffe66d", // yellow
9    "#1a535c", // dark blue
10 ];
11
12 // BAR CHART
13 new Chart(document.getElementById("barChart"), {
14   type: "bar",
15   data: {
16     labels: labels,
17     datasets: [
18       {
19         label: "Students Enrolled",
20         data: dataValues,
21         backgroundColor: colors,
22       },
23     ],
24   },
25   options: {
26     responsive: true,
27     plugins: {
28       legend: {
29         display: true,
30       },
31     },
32   },
33 });
34
35 // PIE CHART
36 new Chart(document.getElementById("pieChart"), {
```

```
35 // PIE CHART
36 new Chart(document.getElementById("pieChart"), {
37   type: "pie",
38   data: {
39     labels: labels,
40     datasets: [
41       {
42         data: dataValues,
43         backgroundColor: colors,
44       },
45     ],
46   },
47   options: {
48     responsive: true,
49   },
50 });
51
```

Output:



Result:

The course enrollment data is effectively visualized using bar and pie charts with a modern and colorful interface.